

End-to-End Clinical Data Management Simulation

Phase III Randomized, Double-Blind, Placebo-Controlled Clinical Trial

XYZ101 — Type 2 Diabetes Mellitus

120-Subject Study Design | 10-Subject Simulated Dataset | Excel, SQL & Power BI

Portfolio Project Report

Educational and portfolio simulation — not professional clinical-trial experience.

1. Executive Summary

This portfolio project demonstrates an end-to-end Clinical Data Management (CDM) workflow using a simulated Phase III randomized, double-blind, placebo-controlled study of XYZ101 in Type 2 Diabetes Mellitus. The planned study design contains 120 subjects; 10 representative simulated subjects were used for hands-on implementation.

The project covers protocol review, data requirements, eCRF design, CDASH mapping, simulated EDC build, edit checks, UAT, data entry, query management, data cleaning, medical coding, external-data reconciliation, data review, simulated database freeze/lock-readiness, SQL validation and Power BI visualization.

The simulated analysis produced a mean Baseline HbA1c of 8.05%, mean Week 12 HbA1c of 7.20%, and mean reduction of 0.85 percentage points. All 10 simulated subjects had a positive HbA1c reduction. These are simulated portfolio outputs, not clinical-trial results.

2. Project Purpose and Objectives

- Translate protocol requirements into structured data requirements and eCRF fields.
- Apply CDASH concepts and consider downstream SDTM mapping.
- Configure and test representative edit checks and UAT scenarios.
- Demonstrate data entry, query management, cleaning, coding, reconciliation and data review.
- Use SQL for independent validation and basic clinical-data analysis.
- Use Power BI to communicate study-level and data-quality metrics.
- Create a truthful, reproducible portfolio demonstration without representing simulated work as professional experience.

3. Study Overview

Item	Definition
Study ID	XYZ101
Phase	III
Design	Randomized, double-blind, placebo-controlled, parallel-group
Population	Type 2 Diabetes Mellitus
Planned sample size	120 subjects
Hands-on dataset	10 simulated subjects
Primary efficacy variable	HbA1c
Key comparison	Baseline vs Week 12
Tools	Excel, MySQL, Power BI
Standards referenced	CDASH and SDTM concepts

4. End-to-End CDM Workflow

1. Protocol review and data requirements definition
2. eCRF design and annotated eCRF/CDASH mapping
3. EDC build/configuration simulation
4. Edit-check configuration and testing

5. User Acceptance Testing (UAT)
6. 10-subject simulated data entry
7. Query management and data cleaning
8. Medical coding
9. External data reconciliation
10. Data review and quality assessment
11. Simulated database freeze / lock-readiness review
12. SQL validation and analysis
13. Power BI dashboard
14. Final project report

5. Documentation and eCRF Design

5.1 Data Requirements and DMP

A data requirements matrix and short Data Management Plan were prepared to define expected study data, forms, review considerations and quality controls.

5.2 eCRF Design

The eCRF design covered demographics, consent, eligibility, medical history, medications, vital signs, laboratory data, efficacy, adverse events, study-drug exposure/compliance, and study completion/early termination.

5.3 Annotated eCRF and CDASH Mapping

Each eCRF form was mapped using Field Label, Project Field Name, CDASH Variable, CDASH Concept, SDTM Domain, SDTM Target and Mapping Notes. CDASH was used as the collection-standard reference and SDTM concepts were considered for downstream standardization.

6. Simulated EDC Build

#	Form
1	Demographics
2	Consent
3	Eligibility
4	Medical History
5	Medications
6	Vital Signs
7	Labs
8	Efficacy
9	Adverse Events
10	Drug Exposure
11	Completion

The Excel-based EDC simulation used 10 representative subjects while retaining the overall 120-subject study design. The simulated records were structured to mirror a clinical database rather than represent real patient data.

7. Edit Checks and UAT

Thirty representative edit checks were configured for uniqueness, requiredness, age, consent, eligibility, medication dates, vital signs, laboratory values, efficacy assessments, adverse events, exposure/compliance and completion logic.

Edit-check testing used test inputs, corrected values and re-testing. Ten representative UAT scenarios were documented and marked as passed in the simulation.

8. Data Entry, Queries and Cleaning

Ten representative subjects were entered. Ten representative queries were logged and resolved, covering eligibility, medical history, medication dates, vital signs, HbA1c, adverse-event dates and compliance.

Metric	Result
Representative queries	10
Resolved queries	10
Open queries at freeze	0
Cleaning records	10
Completed cleaning records	10

9. Medical Coding

Representative medical-history and adverse-event terms were reviewed for coding. MedDRA was used as the simulated dictionary context for medical-event terminology. Medication coding was treated separately; proprietary dictionary codes were not invented.

Area	Records	Status
Medical history	5	Completed
Adverse events	5	Completed
Total representative terms	10	Completed

10. External Data Reconciliation

Ten representative central-laboratory HbA1c records were reconciled between EDC and an external laboratory dataset. One intentional discrepancy was included: Subject 003 had 8.5 in EDC versus 8.6 externally. The simulated process documented verification, EDC correction and resolution.

Metric	Result
Records reconciled	10
Intentional discrepancy	1
Resolved	1
Unresolved at freeze	0

11. Data Review and Database Freeze

Data review covered demographics, consent, eligibility, medical history, medications, vitals, labs, efficacy, adverse events and completion. A simulated freeze checklist confirmed that representative queries, cleaning, coding, reconciliation and review activities were addressed.

Freeze check	Status
Query review	Completed
Data cleaning	Completed
Medical coding	Completed
External reconciliation	Completed
Data review	Completed
Database freeze simulation	Completed

The freeze is a portfolio simulation and does not represent an actual sponsor database lock.

12. SQL Validation and Analysis

12.1 Validation Checks

- Duplicate Subject_ID detection
- Age range validation
- Blood-pressure plausibility checks
- Systolic BP greater-than-diastolic BP check
- Missing laboratory-result check
- HbA1c plausibility check
- Cross-table subject consistency checks
- Duplicate subject/visit check for efficacy

12.2 Efficacy Analysis

Analysis	Simulated output
Subjects analyzed	10
Mean Baseline HbA1c	8.05%
Mean Week 12 HbA1c	7.20%
Mean HbA1c reduction	0.85 percentage points
Subjects with reduction	10/10
Percentage with reduction	100%

The analysis treated Week 12 as a scheduled efficacy assessment, not as End of Treatment. Study completion remained a separate form.

13. Power BI Dashboard

The final dashboard is titled “XYZ101 Clinical Data Management Dashboard” with the subtitle “Phase III Type 2 Diabetes Mellitus — CDM Simulation.”

- Total Subjects — 10
- Subjects With HbA1c Reduction — 10
- Mean HbA1c Reduction — 0.85

- Average HbA1c by Visit — Baseline vs Week 12
- Adverse Events by Severity — Mild vs Moderate
- Study Completion Status — 10 Completed

14. Key Deliverables

1. Protocol
2. Data Requirements Matrix
3. Short DMP
4. eCRF Design Specification
5. Annotated eCRF with CDASH mapping
6. Simulated EDC build
7. Edit Check Specification
8. Edit Check Testing
9. UAT
10. 10-subject simulated data entry
11. Query Log
12. Data Cleaning Log
13. Medical Coding Log
14. External Data Reconciliation Log
15. Data Review
16. Database Freeze simulation
17. SQL Validation and Analysis
18. Power BI Dashboard
19. Final Project Report

15. Skills Demonstrated

Area	Demonstrated capability
Clinical Data Management	eCRF design, edit checks, UAT, queries, cleaning, coding, reconciliation, data review and freeze readiness
CDISC / CDASH	CDASH-based field mapping and SDTM-domain considerations
Excel	Data collection simulation, structured datasets and tracking logs
SQL / MySQL	Validation, joins, aggregation, conditional logic and subject-

	level analysis
Power BI	Data modeling, relationships, measures, KPI cards, charts and dashboard presentation
Data Quality	Discrepancy identification, resolution, reconciliation and review

16. Limitations

- The project is a portfolio simulation and does not represent professional clinical-trial work.
- The study design is 120 subjects, but only 10 representative subjects were entered for hands-on simulation.
- The dataset is fictional and contains no real patient information or confidential sponsor data.
- SQL and Power BI outputs are descriptive portfolio analyses and are not intended for clinical or regulatory decisions.
- Some plausibility ranges and test scenarios are simulation assumptions created to demonstrate data-quality controls.
- Drug-exposure records are representative rather than a complete day-by-day dosing history.

17. Conclusion

This project demonstrates a structured CDM workflow from data-collection design through data review, cleaning, reconciliation, freeze readiness, SQL validation and visualization. It combines CDM concepts with practical Excel, SQL and Power BI skills in a reproducible portfolio format.

All study data and results in this report are simulated and are presented solely for educational and portfolio purposes.